PhantomFlow: Hand-penetrating Sensation using Dual-Sided Vibrotactile Arrays

Team 1:

Soyeong Yang[1], Nakhyeong Lee[2], Cheolsu Kim[3]

Presenter: Kwan Ho Yiu[3], Ryan Lee[3]

1. HCI Tech lab, KAIST MV
2. HRI lab, KAIST ME
3. SRIM lab, KAIST ME

Contents

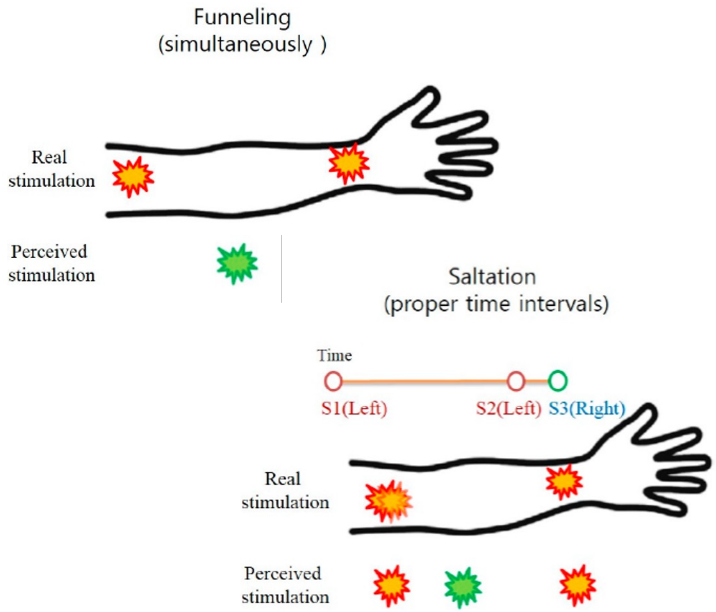
I. Introduction

Introduction

Background & Motivation

An array of vibration motors can create haptic illusions such as **funneling** and **saltation**.

- Most haptic illusions have focused on **horizontal effects** within a 2D plane.
- However, the perception of **depth or inward motion** in haptic feedback remains underexplored.



We extend haptic illusions to a vertical 2D sensation that appears to penetrate the body.

Lee, Hye Jin, et al. "Brain process for perception of the “out of the body” tactile illusion for virtual object interaction." *Sensors* 15.4 (2015): 7913-7932.

Introduction

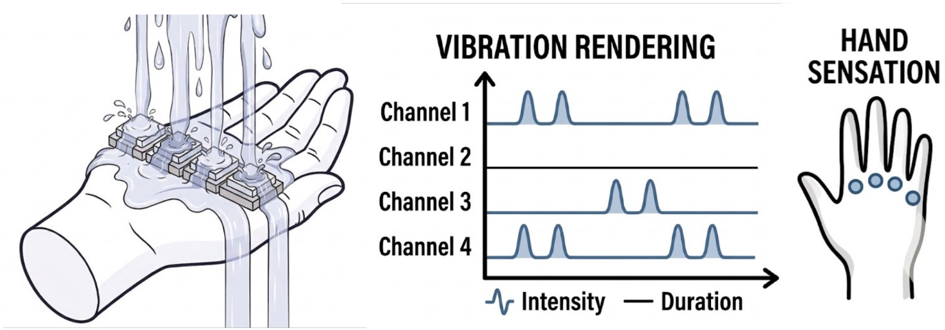
Haptic Scenario

“Can vertical haptic illusions create the sensation of water penetrating the hand?”

<Rendering Goals>

- Continuous droplets are perceived as penetrating the hand at multiple points.
- Stimulation intensity increases with the amount of water.

<Waterfall Scenario Concept>



Introduction

Design Rationale

Selector of LRAs(Linear Resonant Actuators) over ERM

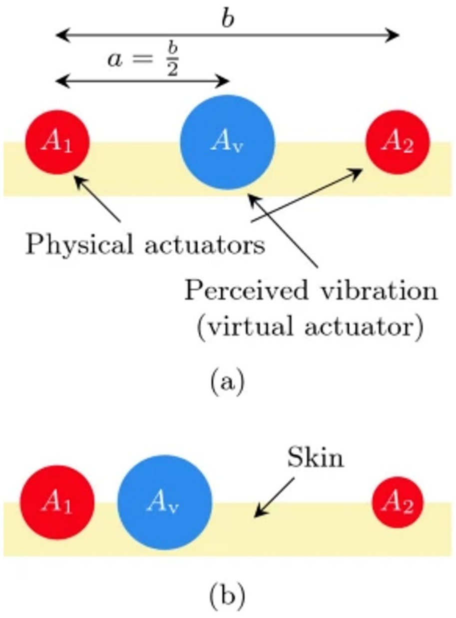
- ERMs couple amplitude and frequency, which distorts sensory blending
- LRAs provide precise amplitude control at a fixed frequency, an essential requirement for generating **seamless Funneling Illusions**

High-Density Actuator Array

- The human palm possesses high tactile sensitivity and spatial resolution
- Dense spatial distribution is required to bridge sensory gaps, preventing jumps

Vertical (Top-Bottom) Arrangement

- Maximizes normal force transmission to the skin surface
- Enhances the **Penetrating Illusion**, simulating deep, structural impacts



Liu et al. "Linear resonator actuator-constructed wearable haptic system with the application of converting remote grinding force to vibratory sensation." *Actuators*. Vol. 12. No. 9. MDPI, 2023.
Seiler et al. "Wearable vibrotactile interface using phantom tactile sensation for human-robot interaction." *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*. Cham: Springer International Publishing, 2020.

Introduction

Design Rationale

Body penetrating sensation

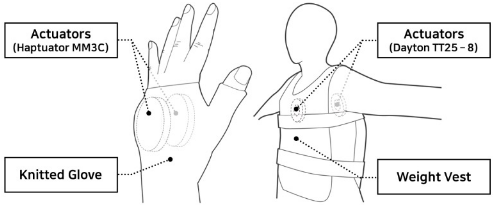


Figure 3. Schematic description of vibrotactile displays for hand (left) and torso (right).

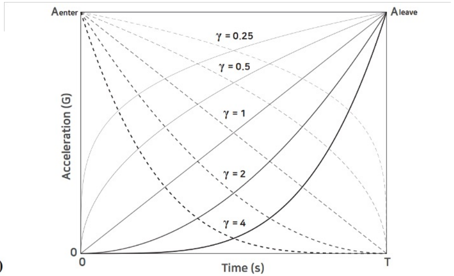


Figure 5. Modulation of vibration amplitudes for different degrees γ of polynomial. Dotted lines represent the amplitude of the actuator for entering stimulus $A_{enter}(freq, body)$. Solid lines are for $A_{leave}(freq, body)$.

Simultaneous exponential intensity control of actuators on both the palm and dorsal sides

Fluid-tactile rendering

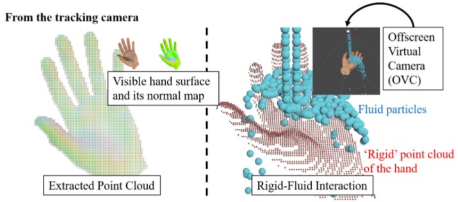


Fig. 3. Point cloud extraction for rigid-fluid interaction. During the simulation, the virtual camera tracks each hand, and the camera viewing direction is the same as the face of ultrasound transducer array to generate a point cloud in the haptic feedback area. The virtual camera captures the depth and normal information in the world space, and the point cloud with normal information is extracted via GPGPU processing.

Fluid-rigid particle simulation incurs high computational cost and long rendering time

Kim, Jinsoo, et al. "Body-penetrating tactile phantom sensations." Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. 2020.
Jang, Jaehyun, and Jinah Park. "SPH fluid tactile rendering for ultrasonic mid-air haptics." *IEEE transactions on haptics* 13.1 (2020): 116-122.

II. System Design

System Design

Preliminary Concept: 1D Linear Array ×2

To generate funneling illusion

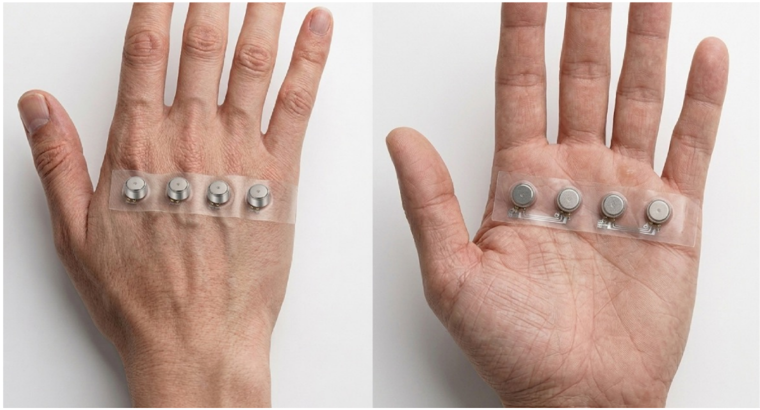
- Gap should be less than **Vibrotactile Spatial Acuity**
- Average palm width ~80mm
- 4 LRAs with ~20mm gap ensure VSA overlap

To generate penetrating illusion

- Apply synchronous stimuli to opposing pairs
- 1:1 Symmetric alignment on dorsum and palm

Reported Vibrotactile Spatial Acuity (VSA)

Palm [1]	Back of the hand[2]
25~30mm	25–50mm



[1] Elsayed, M., et al. (2023). "Understanding Stationary and Moving Direct Skin Vibrotactile Stimulation on the Palm." *CHI Conference on Human Factors in Computing System*
*[2]*Cholewiak, R. W., & Collins, A. A. (2003). "Vibrotactile pattern discrimination and localization at multiple body sites." *Perception & Psychophysics*

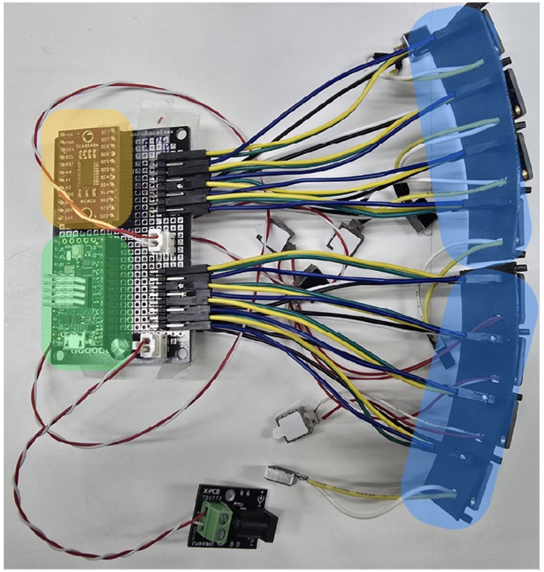
System Design

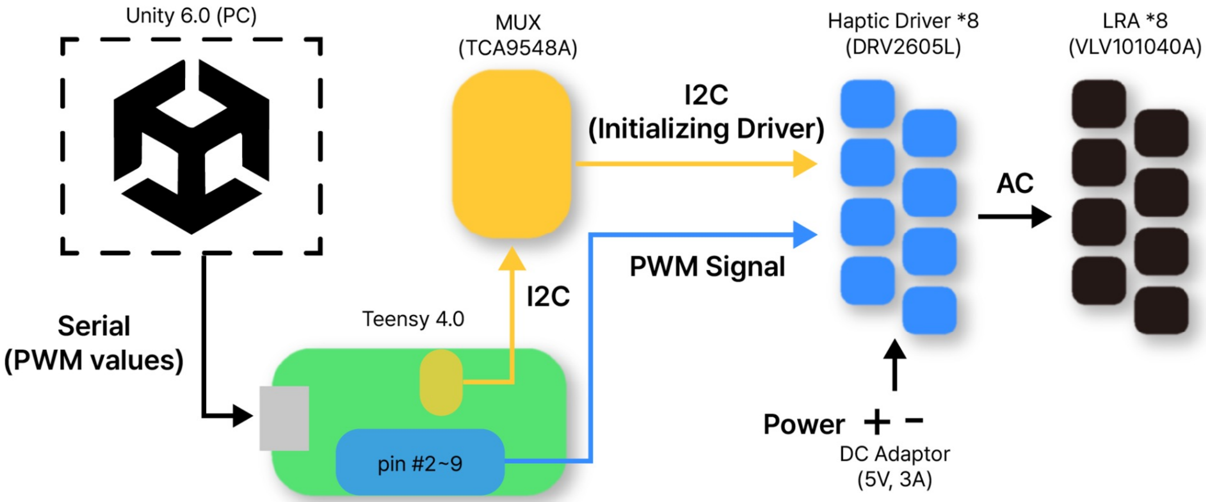
Control Network Layout

Haptic rendering is done in Unity, device simply executes it

Using drivers in a PWM mode, instead of I2C

- Many boards on the hand → Many wires
- **EMI noise can generate serious crosstalk in I2C signal**
- not for a PWM signal



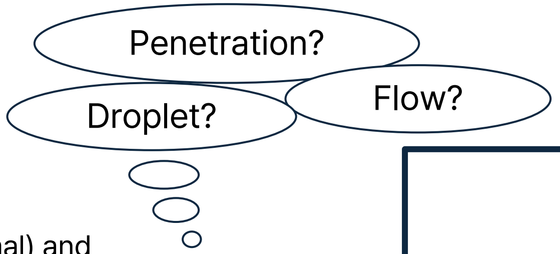


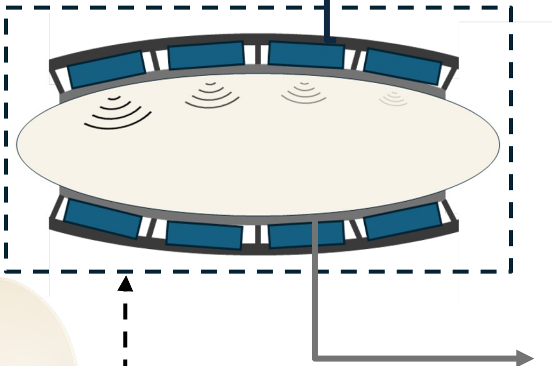
** If initialization using I2C fails, reconnection through unity is possible

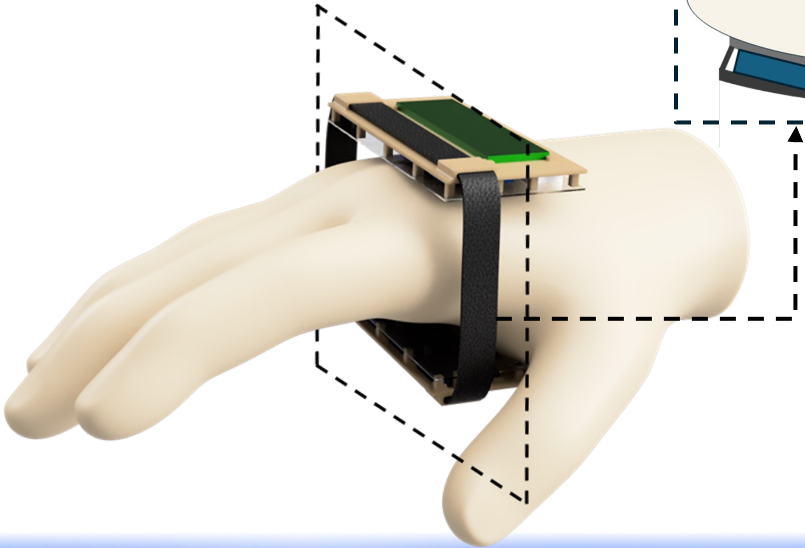
System Design

Hardware Structure

- **Multi-Directional Illusion**:
  Generates both penetrating (normal) and surface-flowing (tangential) sensations.
- **Compact Form Factor**:
  Stacked architecture for seamless wearable integration.







Part 1: TPU Skeleton

Flexible Frame

- **Stable, Normal Positioning:**
  Maximizes haptic transmission efficiency via rigid motor fixation in normal direction.
- **Ergonomic Fit:**
  Conforms to body curvature for stable, continuous contact.

Part 2: Silicone Elastomer film

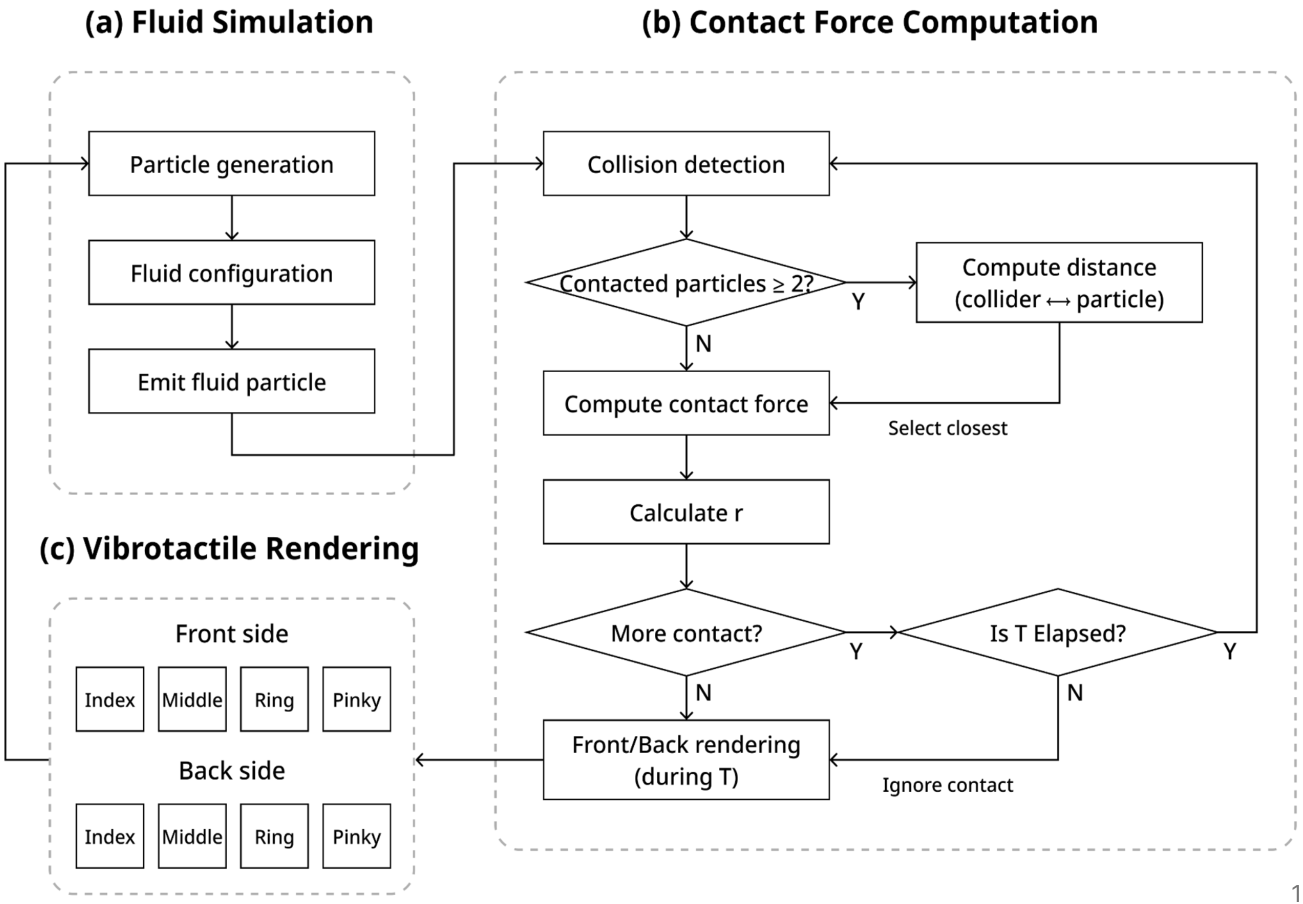
Contact interface layer

- **Vibration Coupling:**
  Enhances actuator-to-skin vibration transfer.
- **Mechanical Low-Pass Filter:**
  Attenuates high-frequency noise for smoother haptic output.
- **Wearable Comfort:**
  Provides soft, safe contact.

III. Haptic Rendering

Haptic Rendering

Rendering Algorithm

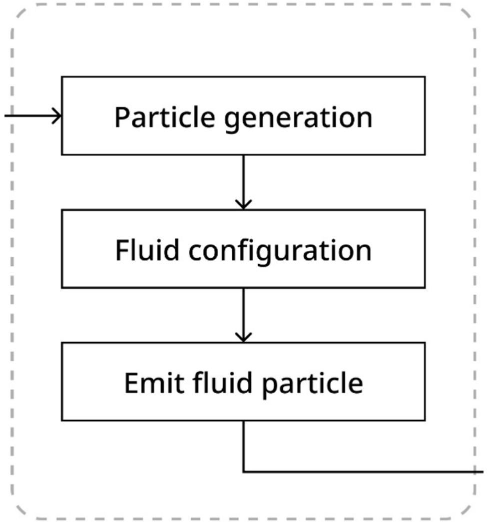


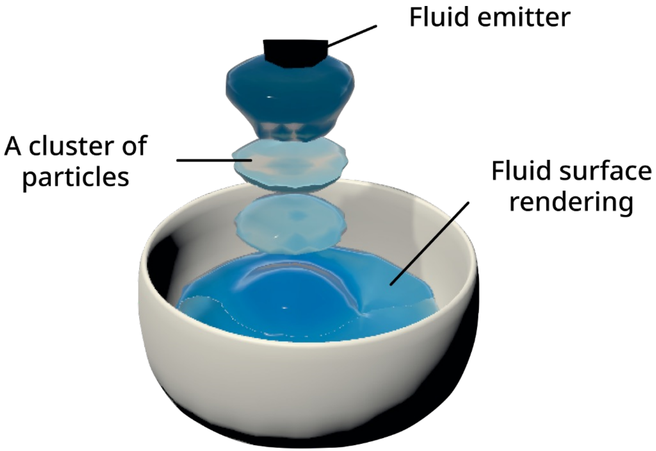
r : Decaying factor (γ)
T: Rendering duration

Haptic Rendering

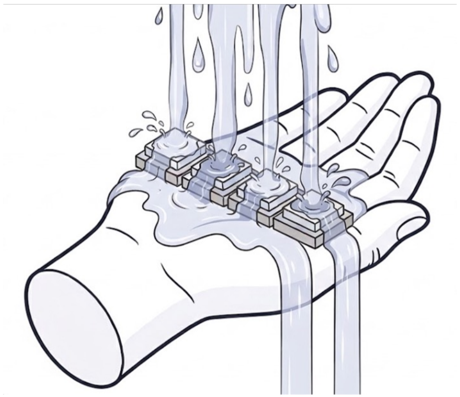
Rendering Algorithm

(a) Fluid Simulation





Obi Fluid
(Unity Asset Store)

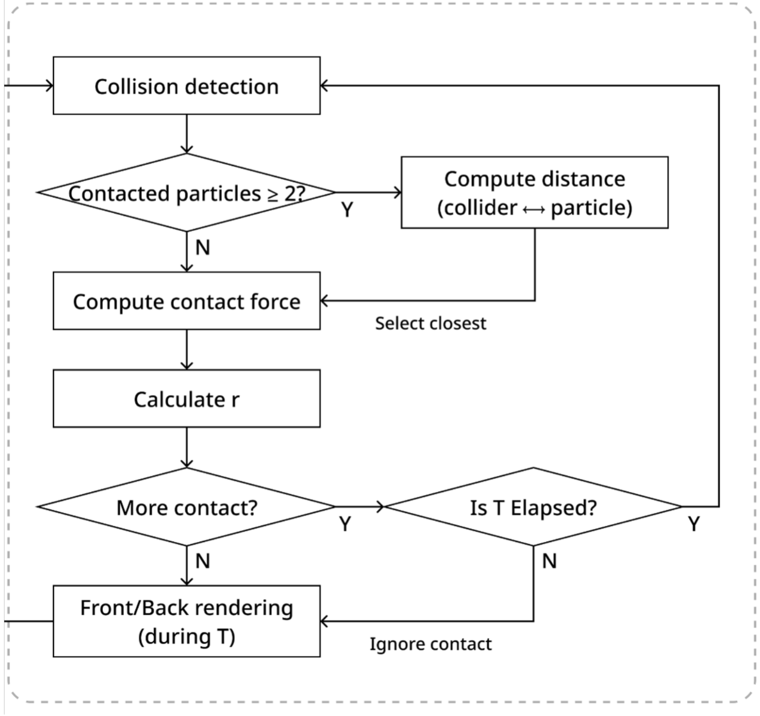


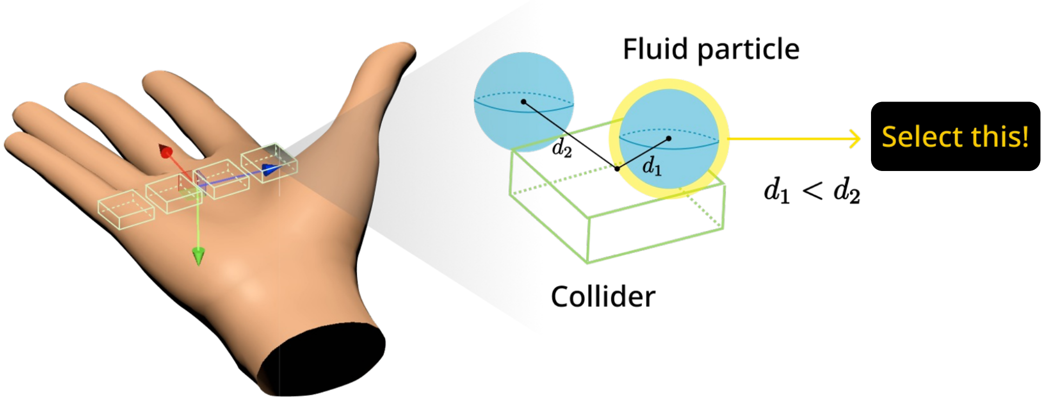
In our system,

Haptic Rendering

Rendering Algorithm

(b) Contact Force Computation





① Decaying gamma (γ)

- Front motor: $\left(1 - \frac{t}{T}\right)^{\gamma}$
- Back motor: $\left(\frac{t}{T}\right)^{\gamma}$

t : current time $(1 \leq \gamma \leq 2)$

② Rendering duration (T)

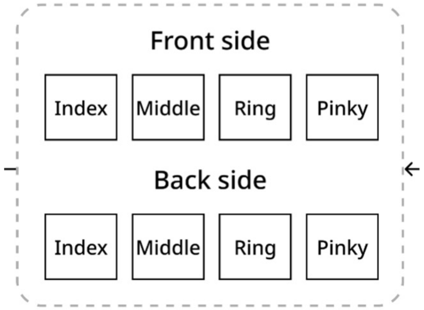
$$T = \sqrt{\frac{2d}{g}}$$

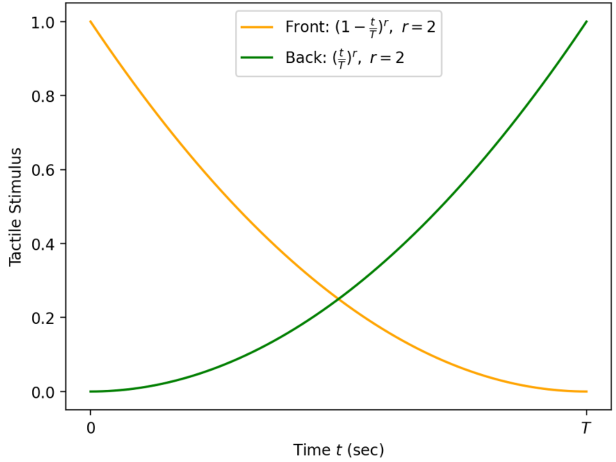
d : distance (palm thickness)
g : gravitational acceleration ($\approx 9.8\ m/s^2$)

Haptic Rendering

Rendering Algorithm

(c) Vibrotactile Rendering



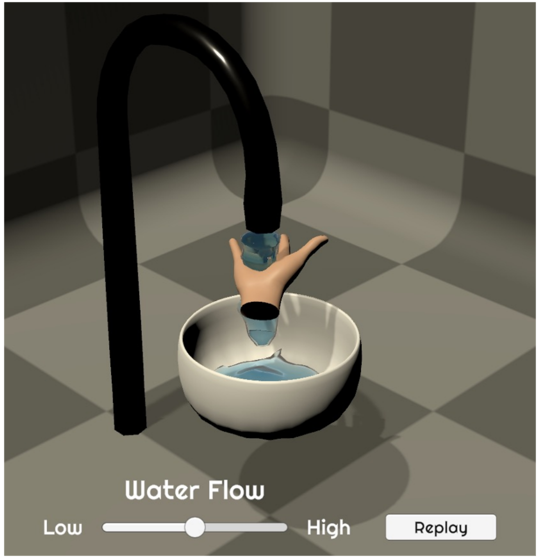


Simultaneous rendering on front and back vibrotactile motors for each finger

Haptic Rendering

Application Scene

Low Intensity



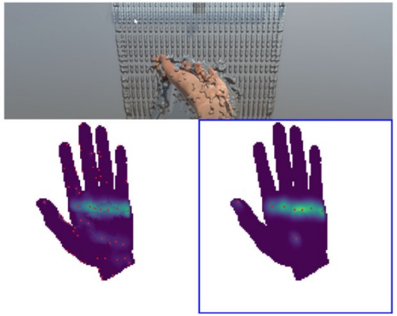
Medium Intensity

High Intensity

Haptic Rendering

Future Work & Scalability

More interactive…



Fluid rendering for different poses [IEEE TOH' 20]

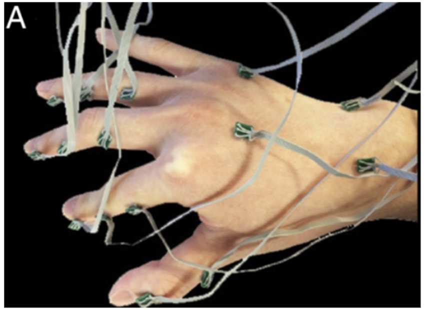
- Incorporate an IMU sensor to reflect **the hand tilt** for more realistic sensation

More haptic sense…

Vibrotactile illusion [CHI EA' 23]

- Refine wiring and structural design for **a slimmer, lighter and more ergonomic build**

More true-to-life…



Vibrotactile propagation [PNAS'16]

- Optimize **actuator placement** based on hand sensitivity
- Determine appropriate **(γ, T)** through perception studies

IV. Demonstration

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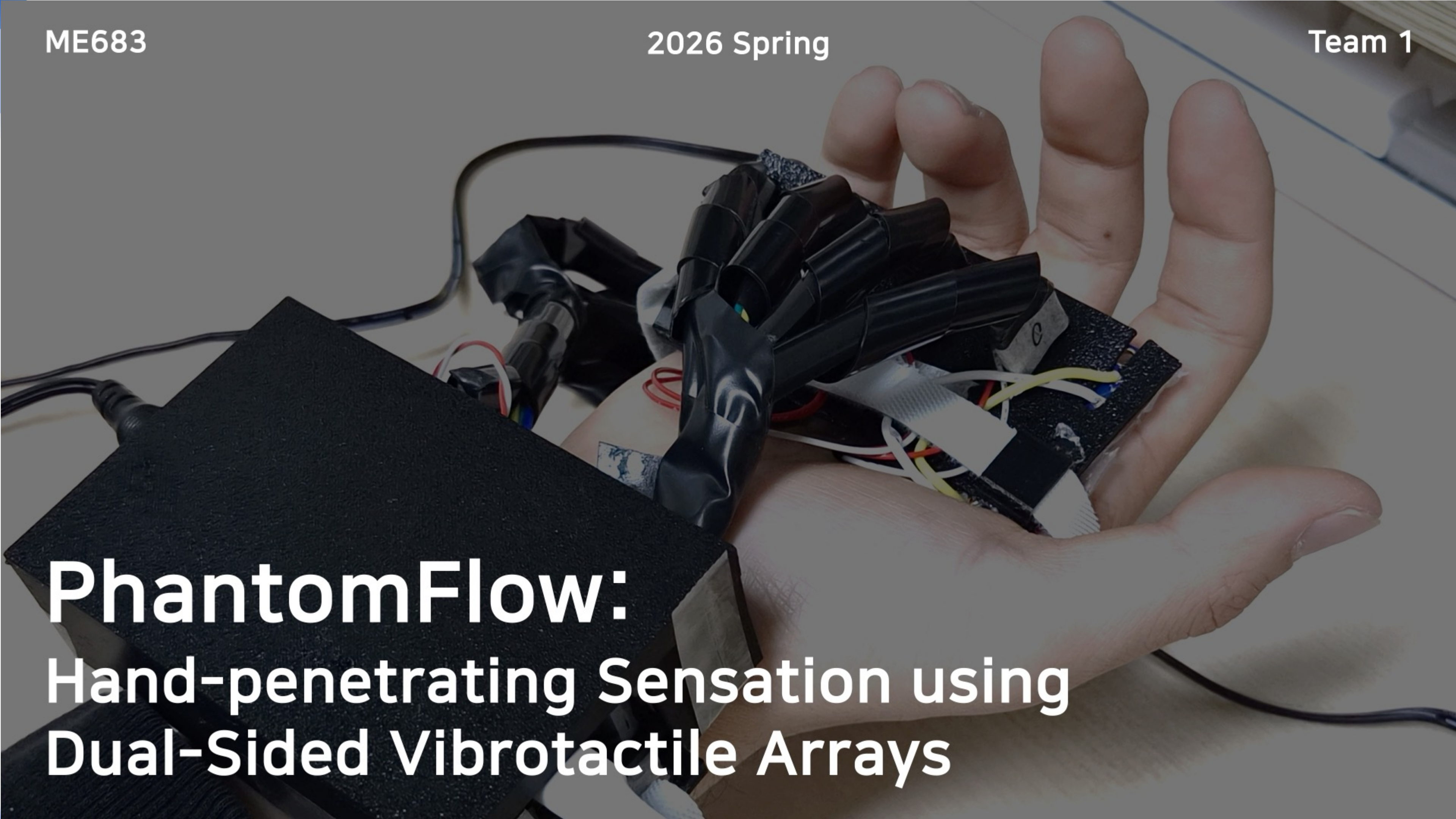

PhantomFlow:

Hand-penetrating Sensation using Dual-Sided Vibrotactile Arrays

Thank you